TASK 5:

Prerequisites:

- MySQL Database installed.
- MySQL Connector jar file
- JDK
- SET PATH FOR JAVA

Installation Steps:

1.

Download MySQL Installer

MySQL 8.0 is the final series with MySQL Installer. As of MySQL 8.1, use a MySQL product's MSI or Zip archive for installation. Read more

2. **Select the file and download**

Select Version:

8.0.45

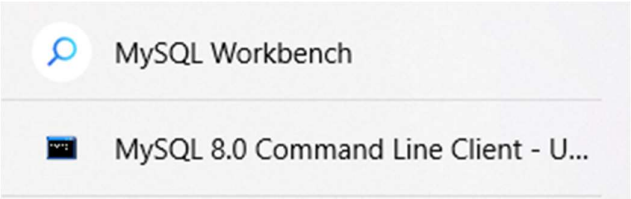
Select Operating System:

Microsoft Windows

Windows (x86, 32-bit), MSI Installer	8.0.45	2.1M	Download
(mysql-installer-web-community-8.0.45.0.msi)	MD5: c221d45c90c003b585b9570edc46a8ad | Signature		

3. **Install**
4. **Provide password (remember it)**
5. **After successful installation , u can see the mysql folder in the pacified path .**

6. **U can also see the mysql command prompt .**



Some of the MYSQL commands:

1. **Show databases; -- displays all the databases available.**
2. **To select particular database to create a tables**
   a. **Use dbname;**
3. **To show tables present in db**
   a. **Show tables;**
4. **To display the schema of the table**
   a. **Describe dbname.tablename;**

CRUD Operations in JDBC with MySQL [task 5]

Step 1. Database Setup in MySQL

- In this step, we create a database and a table where student records will be stored.
- Run the following commands in MySQL Workbench / CLI:

CREATE DATABASE studentdb(last 3 digits of ur roll nub);

USE studentdb5xx;

```
CREATE TABLE students (
   id INT AUTO_INCREMENT PRIMARY KEY,
   name VARCHAR(100) NOT NULL,
   age INT NOT NULL,
   email VARCHAR(150) UNIQUE
);
```

After this u can use java programming to connect to the db and work with the table, to perform operations like Insert, Delete, Display and Update on the table.

To run java program

1. select mysql jar file , store it in the folder where java files are stored.
2. include the jar file in the classpath of the java program to get connected to mysql

java –cp “mysql-connector-j-9.6.0.jar;.” class-name

InsertData.java

```
import java.sql.*;
import java.util.Scanner;

public class InsertData
 {
```

```
   public static void main(String[] args)
{

String url =                                   Port number
"jdbc:mysql://localhost:3307/studentdb5xx";  //
JDBC URL , give ur port number where mysql is
running.
       String userName = "root";  // MySQL
username
       String password = "ur password";  // MySQL
password

try
{
Connection con =
DriverManager.getConnection(url, userName,
password);
System.out.println("Connected to database");
String sql = "INSERT INTO students (id,
name,age,email) VALUES (?, ?, ?, ?)";
PreparedStatement preparedStatement =
con.prepareStatement(sql);
      Scanner sc = new Scanner(System.in);
           System.out.println("Inserting Data into
student table:");
```

```
System.out.println("_____________________________
_____________________");

            System.out.print("Enter student id: ");
            int sid = sc.nextInt();
              sc.nextLine();
            System.out.print("Enter student name: ");
            String sname = sc.nextLine();
            System.out.print("Enter student age: ");
            int sage= sc.nextInt();

              sc.nextLine();
             System.out.print("Enter students email: ");
            String semail = sc.nextLine();
 preparedStatement.setInt(1, sid);     // Sets the first
'?' to the integer sid
            preparedStatement.setString(2, sname); //
Sets the second '?' to the string sname
            preparedStatement.setInt(3, sage);
            preparedStatement.setString(4, semail);

int rowsAffected =
preparedStatement.executeUpdate();

if (rowsAffected > 0) {
```

```
            System.out.println("Record Inserted
Successfully!");
        } else {
            System.out.println("Record insertion
failed.");
        }

    }
catch (SQLException err)
{
        System.out.println("ERROR: " + err);
    }
  }
}
```

DisplayData.java

```
import java.sql.*;

public class DisplayData {
  public static void main(String[] args) {
    try (Connection con =
DriverManager.getConnection("jdbc:mysql://local
host:3307/studentdb", "root", "enter your
password");
        Statement s = con.createStatement()) {
```

```
            ResultSet rs = s.executeQuery("SELECT *
FROM students");
            if (rs != null) {
                System.out.println("id \t name \t age \t
email");

System.out.println("_____________________________
____________________");

                while (rs.next()) {
                    System.out.println(rs.getString("id") +
" \t " + rs.getString("name") + " \t " +
rs.getString("age") + rs.getString("email"));

System.out.println("_____________________________
____________________");
                }
            }

        } catch (SQLException err) {
            System.out.println("ERROR: " + err);
        }
    }
}
```

DeleteData.java

```
import java.sql.*;
import java.util.Scanner;

public class DeleteData {
   public static void main(String[] args)
{
       try (Connection con =
DriverManager.getConnection("jdbc:mysql://local
host:3307/studentdb5xx", "root", "your
password”);
            Statement s = con.createStatement()) {

           Scanner sc = new Scanner(System.in);
           System.out.println("Delete Data from
student table:");

System.out.println("_____________________________
_____________________");

           System.out.print("Enter student id: ");
           int sid = sc.nextInt();

           String deleteQuery = "DELETE FROM
students WHERE id=" + sid;
           s.executeUpdate(deleteQuery);
```

```
            System.out.println("Data deleted
successfully");

        } catch (SQLException err) {
            System.out.println("ERROR: " + err);
        }
    }
}
```

updateData.java

```
import java.sql.*;
import java.util.Scanner;

public class UpdateData {
    public static void main(String[] args)
{
try (Connection con =
DriverManager.getConnection("jdbc:mysql://local
host:3306/studentdb5xx", "root", "your
password");
            Statement s = con.createStatement())

{

            Scanner sc = new Scanner(System.in);
```

```
        System.out.println("Update Data in student
table:");

System.out.println("_________________________
___________________");

        System.out.print("Enter student id: ");
        int sid = sc.nextInt();
        System.out.print("Enter student name: ");
        String sname = sc.next();
System.out.print("Enter student age: ");
        int sage = sc.nextInt();
        System.out.print("Enter student email: ");
        String semail = sc.next();

        String updateQuery = "UPDATE students
SET name='" + sname + "', age='" + sage +
"',email='" + semail + "'  WHERE id=" + sid;
        s.executeUpdate(updateQuery);

        System.out.println("Data updated
successfully");

      } catch (SQLException err) {
        System.out.println("ERROR: " + err);
      }
    }
```

}

Out pattern [for all the java programs, screen shots printed and properly pasted both in observation and records]

for e.g.- Insert operation : command prompt output

```
D:\MGIT\2026 II-II sem\NODEJS-Lab\lab programs\task5>javac InsertData.java

D:\MGIT\2026 II-II sem\NODEJS-Lab\lab programs\task5>java  -cp "mysql-connector-j-9.6.0.jar;." InsertData
Connected to database
Inserting Data into student table:
_____________________________________
Enter student id: 6
Enter student name: xxx
Enter student age: 17
Enter students email: xxx@gmil.com
Record Inserted Successfully!
```

mysql output

```
mysql> select * from students;
+----+-------+-----+--------------------+
| id | name  | age | email              |
+----+-------+-----+--------------------+
|  1 | john  |  25 | john@example.com   |
|  2 | arvin |  16 | arvin@example.com  |
|  3 | aaron |  18 | aaron@gmail.com    |
|  5 | rahul |  19 | rahul@gmail.com    |
|  6 | xxx   |  17 | xxx@gmil.com       |
+----+-------+-----+--------------------+
5 rows in set (0.00 sec)
```